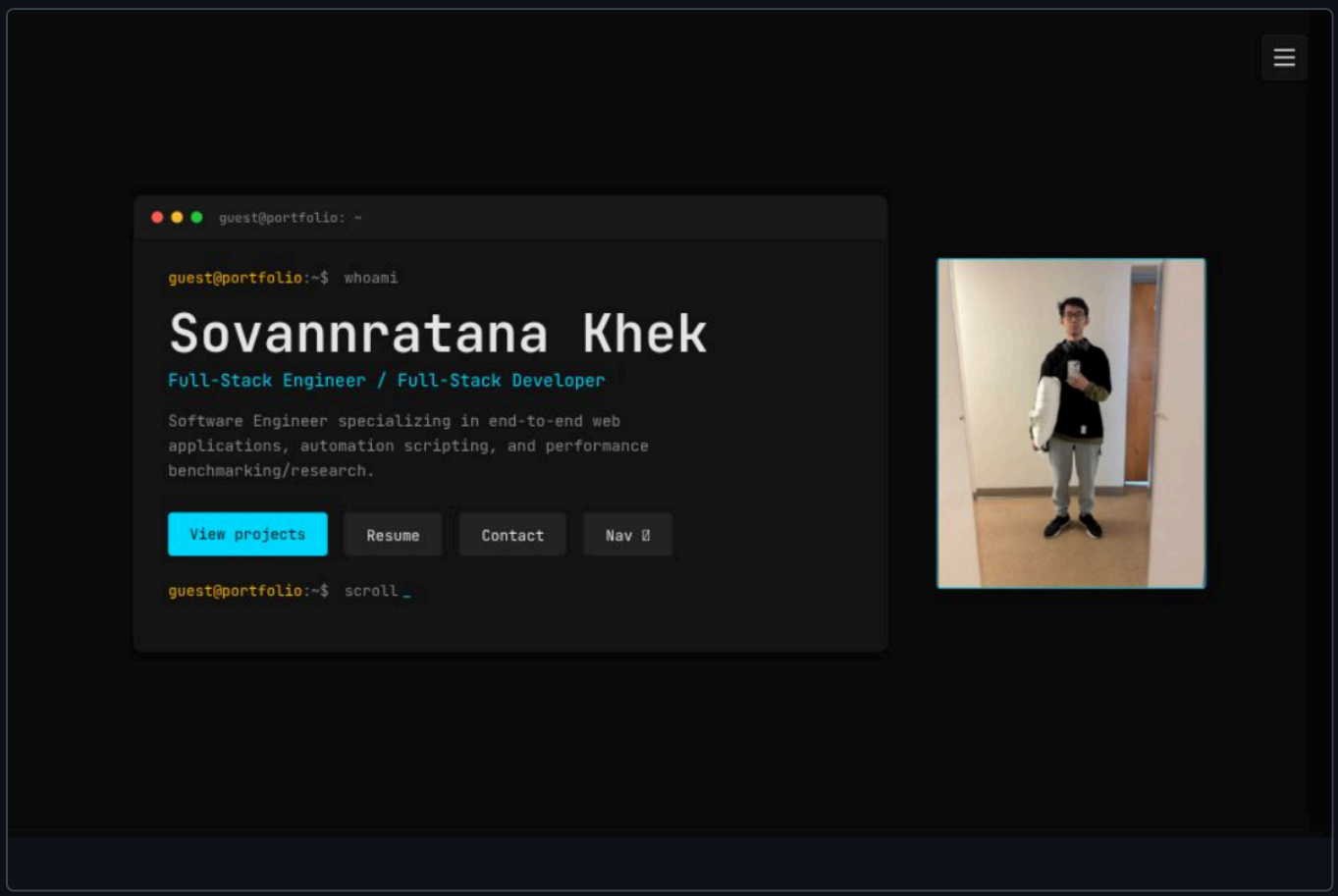
Ratana Khek | Portfolio

A console-themed developer portfolio. Instead of a static list of projects, each one runs as a scripted demo inside an on-page terminal, with interactive previews embedded in the output.

Live site: https://ratana-portfolio.vercel.app/



Overview

The site is a single page that mixes a traditional portfolio (about, experience, education, contact) with one distinctive element: a console. Picking a project types out a short session in the terminal, then opens a preview built for that project. The console is one section of the site, not the whole interface, so a visitor can skim it like any other portfolio.

Features

- **Projects:** a scrollable grid of terminal-style cards. Clicking a card runs that project's script in the console, and clicking a tech tag jumps to that skill.
- **Console:** commands are typed out character by character, followed by output lines and embedded previews:
    - a benchmark table and research paper preview for HNSW insertion research

    - a pipeline diagram, worked example and demo video for a Huffman + Hamming error-correction pipeline
    - a playable digital lock, ported from Python to JavaScript, with an in-browser break-in simulation
    - a feature breakdown and recorded demo for a hackathon AI coach
- **Skills:** grouped skill pills with a terminal-style detail panel. The "used in" project links are derived from each project's tech list, so they can't drift out of date.
- **Education:** a `git log`-style timeline of milestones.
- **Experience:** a `ps aux`-style process list with expandable rows.
- **Navigation:** a slide-in menu with scroll-spy that highlights the current section.
- **Contact:** a working message form (Web3Forms) with browser autofill and a honeypot field to filter bots, plus links to email, GitHub, LinkedIn and a resume download.

Tech stack

Area	Choice
Framework	React with Vite
Styling	CSS Modules and CSS custom properties (no UI library)
State	React Context, composed from small custom hooks
Fonts	JetBrains Mono and Inter
Contact form	Web3Forms (static-site friendly, no backend)

How it's organized

The site is data-driven. Content lives in plain data files and components render it.

- `data/projects.jsx` holds every project: title, blurb, tech list, and the `script` for its console demo. Each project owns its script, so there is no second file to keep in sync.
- `data/skills.js` holds skill names, categories and blurbs. Which projects use a skill is computed from the projects' `tech` arrays and never stored twice.
- `data/education.js` and `data/experience.js` feed the timeline and the process list.
- `context/PortfolioProvider.jsx` is a single provider that composes `useConsoleState`, `useSkillsState` and `useNavState`. Components that share state (the project cards, the console, the skills panel and the nav) read from it.

A console script is an array of steps:

```
script: [
  { type: 'command', text: 'cd Projects/example' },
  { type: 'output',  text: 'Some output line' },
  { type: 'element', content: <ExamplePreview /> },
  { type: 'action',  run: () => window.open('...', '_blank') },
]
```

Running locally

```
npm install
npm run dev
```

The contact form needs a Web3Forms access key. Create a `.env` file in the project root:

```
VITE_WEB3FORMS_KEY=your-access-key
```

Restart the dev server after creating or changing `.env`. To build for production, run `npm run build`, and set the same variable in your hosting provider's environment settings.

Design decisions

- **Hybrid, not all-console.** A fully terminal site is fun but slow to skim, so the console is one section and the rest is conventional.
- **Curated over exhaustive.** The site shows a handful of well-documented projects. The full set lives on GitHub.
- **Single source of truth.** Anything derivable (such as skill-to-project links) is computed, not duplicated.
- **Real data in previews.** The example outputs shown in project previews come from actually running the code, not from invented numbers.

Notes

- The hackathon project (Oracle Lens) no longer runs because its Riot API and AWS Bedrock access expired. The site preserves it with a recorded demo and links to the original source and Devpost page.

Contact

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